Evaluation Rubric: DS Final Project

A Comparative Analysis of Machine Learning Architectures for Satellite-Based Surface Monitoring

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Criteria	Excellent (9–10 pts)	Proficient (6–8 pts)	Developing (2–5 pts)	Unacceptable (0–1 pts)
Data Extraction & Formatting	Successfully exports a **TSV** file containing Latitude, Longitude, ≥ 8 spectral bands, and correct categorical class labels. High-quality extraction via scripts (Python/R).	Exports a TSV file with minor omissions (e.g., missing 1–2 spectral bands or minor coordinate format issues).	Wrong file format used (e.g., CSV instead of TSV) or critical columns missing. Script execution errors.	No dataset or data extraction script provided.
Model Implementation	Correctly implements, trains, and documents **all five required architectures**: DT, SVM, ANN, KNN, and NB.	Implements 4 out of 5 required models, or models are implemented with minor technical execution flaws.	Implements 2–3 models only, or models are poorly configured with no regard for their structural logic.	Fails to implement the machine learning models.
Hyperparameter Optimization (HPO)	Demonstrates an **exhaustive HPO process** for all algorithms; settings are systematically adjusted to maximize performance.	HPO is attempted for all models, but the optimization process lacks depth or systematic testing.	HPO is only performed on a couple of models, or parameters are modified randomly without tuning logic.	No hyperparameter optimization is performed.
Performance Evaluation & Synthesis	Generates precise Confusion Matrices and correctly computes **Overall Accuracy, Kappa, Precision, Recall, and F1-Score**. Synthesizes findings into a clear direct-comparison table.	Computes performance metrics, but has minor calculation errors or misses one key metric. Comparison table is present but hard to read.	Misses multiple evaluation metrics or fails to create a unified synthesis comparison table.	Evaluation metrics and comparison tables are entirely missing.
Raster Prediction & Inference	Successfully deploys the optimal model to run full-scene inference on the original multidimensional raster stack, exporting a georeferenced **GeoTIFF**.	Generates the GeoTIFF inference raster, but contains minor spatial alignment issues or projection hitches.	The output raster is corrupt, lacks spatial georeferencing, or fails to map integer codes to correct categories.	Fails to generate or export the final prediction raster.
Spatial Consistency & GIS Visualization	Perfect retention of original CRS and affine parameters. High-quality discrete color symbology in GIS and accurate class area quantification (ha/km^2).	CRS/affine parameters align, but GIS visualization symbology is unclear or area statistics contain minor calculation errors.	The final map lacks spatial consistency with original imagery, or area statistics are missing.	No GIS visualization or area quantification provided.
Technical Mastery & Readiness	**Randomly selected presenters** demonstrate absolute mastery of the code, algorithms, and workflows, proving collective team knowledge.	Selected presenters understand the project well but struggle to answer deep technical questions about specific model mechanics.	Selected presenters show a clear lack of familiarity with parts of the workflow, indicating uneven team workload distribution.	Presenters are unable to explain the code, methodology, or project architecture.
Communication & Time Management	Professional slide design. High clarity of speech and seamless delivery. Presentation is comprehensive and strictly **does not exceed 20 minutes**.	Presentation is clear but slightly unpolished. Timing is marginally over or under the 20-minute threshold.	Slides are chaotic or text-heavy. Presenters significantly breach the time limit or fail to convey core concepts clearly.	Presentation is unstructured, unprofessional, or completely misses the core project constraints.